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Auditable Reconstruction of Field-Scale Well Construction Timelines from Daily Drilling Reports: The Volve Field Open Dataset

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Abstract

Daily Drilling Reports (DDRs) record how wells were drilled, completed and abandoned, but reconstructing a field-wide construction history requires reconciliation of narrative descriptions, structured fields and wellbore designations. This study reconstructs the available construction history of the Equinor Volve dataset from 1,759 WITSML DDRs covering 26 wellbores in 11 well families.

A reproducible pipeline extracts the source XML, identifies placeholder values and assigns operational labels using a deterministic, priority-ordered classifier. Reconstructed depths, dates and sidetrack relationships are compared with Norwegian Offshore Directorate (NOD) records. Separate audits examine construction content in unclassified reports and agreement between assigned phases and the author's engineering interpretation.

The classifier assigns a phase to 1,303 reports, giving 74.1% coverage. A stratified audit of 50 unclassified narratives identifies additional construction content corresponding to approximately 198 reports. In a second audit of 52 classified reports, agreement with the author's independently assigned labels is 61.0% after weighting by phase population. This measures agreement with engineering interpretation, not accuracy against a reference standard. Reconstructed maximum measured depth agrees with NOD total depth within 0.5 m for 19 of 21 name-matched wellbores; three additional technical sidetracks are matched using depth. The reconstruction also identifies erroneous or inherited spud dates, 515 placeholder field values, and differences between archive and register representations. Sidetrack departure depths or intervals are interpreted for all 15 sidetracks, with evidence strength recorded explicitly.

The published output comprises a 1,421-record depth–time–phase dataset, supporting tables and reproducible code. The study demonstrates the value of combining operational narratives with structured records while exposing the limitations of single-report phase classification. The reconstruction supports records-based screening of legacy wells, but does not establish well integrity or justify direct use of phase counts as operational durations.

1 Introduction

Daily Drilling Reports (DDRs) are the primary operational record of a well construction campaign. Each report logs depth progress, casing and cementing operations, formation encounters and non-productive time. In the WITSML (Wellsite Information Transfer Standard Markup Language) exchange format (Energistics n.d.), a DDR holds structured fields such as measured depth, spud date and casing depths, together with a free-text narrative of the preceding 24 hours of operations. Turning the archive of a field into a construction timeline means assigning every report to a construction phase, and the inconsistent terminology, abbreviations and narrative text that accumulate over a field's multi-decade life make that difficult. This paper reconstructs the timeline of the Equinor Volve field from 1,759 WITSML DDRs covering 26 wellbores. A deterministic, priority-ordered rule-based classifier reads the narrative of every report. A stratified sample of the unclassified remainder shows what the classifier misses, and total depth and spud date are compared for every wellbore with an independent national register, the NOD FactPages (NOD 2026a), and the unmatched and unresolved cases are reported. The data-quality problems that any other user of the archive will meet are documented along the way. A reader with this text, the openly available Volve dataset (Equinor 2018) and the accompanying repository should be able to reproduce every reported figure.

2 Background and Related Work

Several studies have extracted structured information from the free text of drilling reports. Sidahmed et al. (2015) built a dictionary of rig terms and tracked weighted term frequencies in the reports of three wells to flag recurring operational concerns, and Priyadarshy et al. (2017) presented a framework for predicting the causes of non-productive time from unstructured reports. Hoffimann et al. (2018)

applied deep natural language processing to 9,670 daily reports from a field of 303 wells and classified each sentence as an event, a symptom or an action; they noted that technical symbols, abbreviations, misspellings and incomplete sentences make the text difficult to process. Zhang et al. (2022) built a rule-based detector for five unscheduled events (stuck pipe, lost circulation, water influx, gas influx and H2S) in about 128,000 free-text remarks from daily mud reports. The detector was precise but missed events, and its accuracy fell to 65.18% on a larger test set; a hybrid that used its output to train a weakly supervised deep learning model improved detection by more than 20 percentage points.

Karnot et al. (2025) combine change-point detection, semi-supervised natural language processing and sequence mining for root-cause analysis of non-productive time, applied to 701 DDRs from 10 wells; their Dataset section states that "all analyses used synthetic datasets constructed to emulate typical drilling records", and their proof-of-concept sequence-mining section states that "rigorous validation and quantitative performance assessment are left for future work". Their abstract and conclusions nevertheless report an average 18.5% reduction in non-productive time in field applications.

The Volve dataset is an established benchmark for drilling analytics. Tunkiel et al. (2021) derived a reference dataset for rate-of-penetration benchmarking from its drilling data. To the author's knowledge, no validated field-wide construction timeline from the Volve DDR narratives has been published.

Published DDR work relies on proprietary or synthetic data, which prevents others from reproducing or extending its results. The present work (1) uses a deterministic, auditable classifier that trades model sophistication for transparency of every decision, (2) works exclusively with a real, openly licensed archive, and (3) pairs classification with a stratified audit of the unclassified residual, an agreement audit of classified reports against the author's labels, and a comparison of total depth and spud date with a national register for the complete wellbore population. The audit in this paper measures the weakness that Zhang et al. (2022) report, the events a rule-based classifier misses, directly on the unclassified reports.

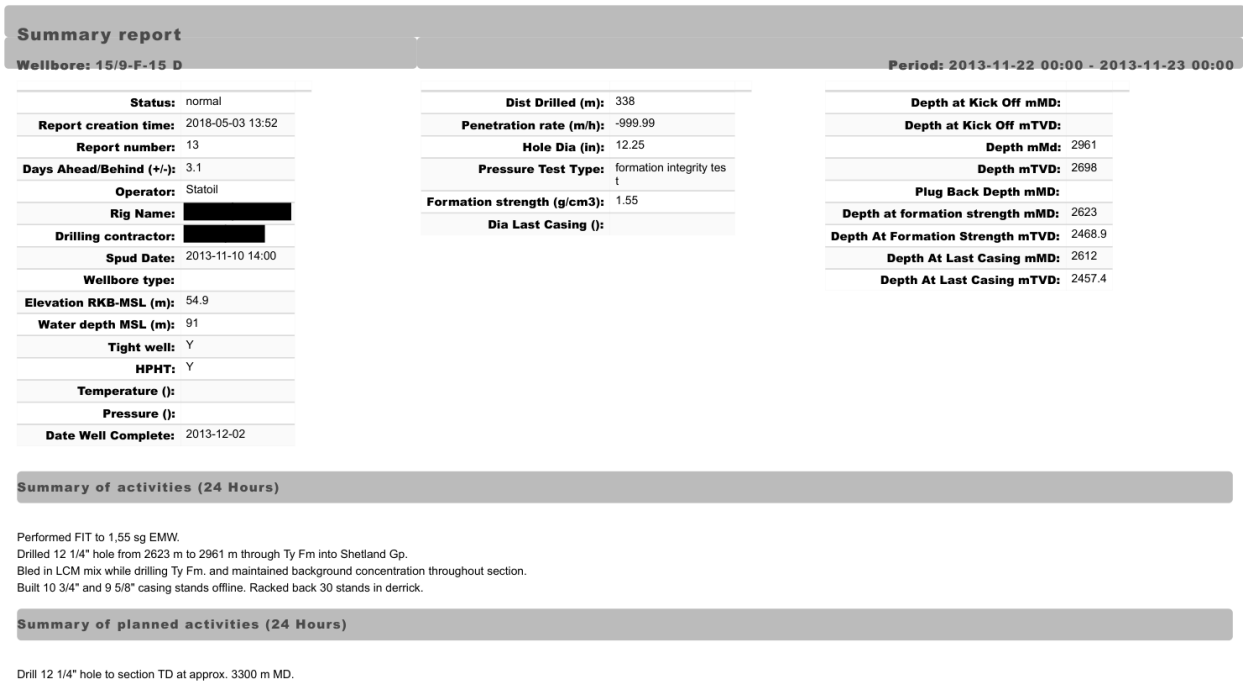
3 Data and Methods

3.1 Data Source and Extraction.

The 1,759 WITSML 1.4.0.0 XML files of the Volve dataset (21,315,879 bytes, each recorded with a SHA-256 hash) were extracted into a single table with one row per DDR and no duplicate wellbore and date pairs. The rows cover the exploration well 15/9-19 and the platform wells 15/9-F-1, F-4, F-5, F-7, F-9, F-10, F-11, F-12, F-14 and F-15, which make up 26 wellbores in 11 well families. The columns include the source filename, the well and wellbore identifiers, the report date, the spud date (WITSML dTimSpud), measured depth (md_m), kick-off depth (md_kickoff_m), last casing depth (md_csg_last_m), and the free-text 24-hour narrative (sum24Hr). Every field was kept as a raw string to keep placeholder values visible, and the published code regenerates the table from the raw files.

The md_m field is stored as a string and contains two placeholders, the literal '-999.99' (a missing-value sentinel) and the literal '0'. Both were excluded and the remaining values cast to a numeric type before any depth arithmetic.

Fig. 1 shows a real construction report as a page of the original PDF version of the Daily Drilling Report: the structured fields that the pipeline uses and the free-text summary that the classifier reads. The same report is stored in the archive as file 15_9_F_15_D_2013_11_23.xml (Equinor Volve dataset, Equinor 2018); its summary has 51 words, and its kick-off field is empty, as in most reports.



Summary report

Wellbore: 15/9-F-15 D

Period: 2013-11-22 00:00 - 2013-11-23 00:00

Status:	normal
Report creation time:	2018-05-03 13:52
Report number:	13
Days Ahead/Behind (+/-):	3.1
Operator:	Statoil
Rig Name:	[redacted]
Drilling contractor:	[redacted]
Spud Date:	2013-11-10 14:00
Wellbore type:	
Elevation RKB-MSL (m):	54.9
Water depth MSL (m):	91
Tight well:	Y
HPHT:	Y
Temperature ():	
Pressure ():	
Date Well Complete:	2013-12-02

Dist Drilled (m):	338
Penetration rate (m/h):	-999.99
Hole Dia (in):	12.25
Pressure Test Type:	formation integrity test
Formation strength (g/cm3):	1.55
Dia Last Casing ():	

Depth at Kick Off mMD:	
Depth at Kick Off mTVD:	
Depth mMd:	2961
Depth mTVD:	2698
Plug Back Depth mMD:	
Depth at formation strength mMD:	2623
Depth At Formation Strength mTVD:	2468.9
Depth At Last Casing mMD:	2612
Depth At Last Casing mTVD:	2457.4

Summary of activities (24 Hours)

Performed FIT to 1,55 sg EMW.
Drilled 12 1/4" hole from 2623 m to 2961 m through Ty Fm into Shetland Gp.
Bled in LCM mix while drilling Ty Fm. and maintained background concentration throughout section.
Built 10 3/4" and 9 5/8" casing stands offline. Racked back 30 stands in derrick.

Summary of planned activities (24 Hours)

Drill 12 1/4" hole to section TD at approx. 3300 m MD.

Fig. 1: Page of the original Daily Drilling Report for 15/9-F-15 D, 23 November 2013 (Equinor Volve dataset, PDF version).

3.2 Wellbore Designation.

Norwegian wellbore names follow the designation guidelines of the national regulator (NOD 2025). The suffix "S" marks an exploration wellbore planned to be deviated, letters A, B and so on mark sidetracks, and "R" marks a re-entered exploration wellbore. A technical sidetrack, drilled to bypass a mechanical problem such as stuck pipe, is designated T2 for the first and T3 for the next. The Volve WITSML archive uses the technical-sidetrack designations 15/9-19 BT2, 15/9-19 ST2 and 15/9-F-11 T2, and none of them appears as a separate entry in the NOD register for the field. This one convention explains every naming mismatch between the two registers (Section 5.5).

3.3 Phase Classification.

The classifier assigns the sum24Hr narrative of each DDR to one of 13 phases: Spud/Conductor, Surface hole and casing, Intermediate drilling, Intermediate casing, 12¼ in. and smaller hole, Liner/Casing run, Cementing, Completion, P&A, Logging/Testing, Workover, Waiting/Downtime and Rig move/Setup. Each phase has a fixed priority from 1 (highest) to 13 (lowest), in the order listed, and a set of case-insensitive regular expressions. A DDR takes the highest-priority phase with a matching pattern, and a narrative that matches no pattern stays Unclassified. Fraction characters such as ½ and ¼, routine in the mixed units of drilling reports, are stored as single special characters that a pattern written as "1/2" does not match. They are converted to that form before matching. The phase names refer to the hole and casing sizes that each pattern group matches. On Volve platform wells the 12¼ in. hole usually ends above the reservoir (Hugin Formation); in 15/9-F-4, for example, the 12¼ in. section ended at 2,770 m MD and pressure samples were taken in the Hugin Formation at 3,417.5 m MD, in the 8½ in. section. Every regular expression is listed in the repository file src/classifier_v3.py.

Table 1 gives the 13 phases in priority order with their hole and casing sizes and example trigger words from the pattern table in the repository (outputs/stage4_priority_rules.csv). Each phase has further patterns that the table does not show.

Table 1: Phases in priority order with representative triggers

Priority	Phase	Hole and casing sizes (in.)	Example trigger words
1	Spud/Conductor	36, 30	spud, tagged sea bottom
2	Surface hole and casing	26, 20	(size only)
3	Intermediate drilling	17½	(size only)
4	Intermediate casing	13⅜, 10¾, 14	(size only)
5	12¼ in. and smaller hole	12¼, 8½, 6	cut core, coring
6	Liner/Casing run	9⅝, 7	liner, surface tieback
7	Cementing	none	cement, cemented, cementing, bumped plug
8	Completion	none	completion, production packer, packer, tubing

Priority	Phase	Hole and casing sizes (in.)	Example trigger words
9	P&A	none	proprietary plug names, P&A, abandon
10	Logging/Testing	none	logging, LWD, MWD, VSP
11	Workover	none	workover, coiled tubing, sidetrack, whipstock
12	Waiting/Downtime	none	WOW, waiting on weather, rig maintenance, general maintenance
13	Rig move/Setup	none	skidded rig, positioned rig, mooring, ballast

A report takes the highest-priority phase with a match, and a report with no match is Unclassified. The triggers shown are simplified: the actual expressions impose context and distance limits, and all of them are in src/classifier_v3.py. Both casing sizes in the table occur in the reports: 14 in. in 31 DDRs (15/9-F-11 B, 15/9-F-11 T2, 15/9-F-12, 15/9-F-14) and 10¾ in. in 104 DDRs (11 wellbores), 68 of which also mention a tie-back or the combined 9⅝ x 10¾ in. string.

3.4 Stratified Audit of the Unclassified Residual.

The audit estimates how many unclassified DDRs describe construction operations that the pattern set failed to catch. A leak check first reapplied the complete pattern set to every unclassified record and found no record that should have matched. Every unclassified record was then assigned to one of eleven descriptive categories, for example BOP and riser handling, generic trips, production monitoring and empty records, using patterns built from inspection of the unclassified population. BOP and marine riser handling is the largest category (147 of 456 records, 32.2%). The 73 empty records have a sum24Hr field that holds the literal string "None" (Section 4). These records contain no operational narrative in the sum24Hr field for the classifier to interpret. They were excluded from the sampling frame, leaving 383 records.

A stratified sample of 50 records was drawn from the 383. The unclassified reports were split into two strata, an early construction group (reports within 730 days of the wellbore's first report, 296 records) and a later production group (reports more than 730 days after it, 87 records), and a random sample was drawn from each in proportion to its size, 39 records from the construction group and 11 from the production group. The random draw is recorded in the repository and allows the same sample to be drawn again. The author read each sampled report and labelled it construction if it described work that builds the well (drilling ahead, coring, running or cementing casing, setting plugs, fishing or completion). Reports that described only BOP or riser handling, tripping, waiting or production were labelled non-construction unless they also named a hole or casing size.

3.5 External Validation Against NOD FactPages.

Total depth and spud date were compared with the NOD FactPages (NOD 2026a). The Table View export was filtered to the Volve wellbores and matched by exact name wherever possible. Three WITSML wellbores, all technical sidetracks, were matched by depth (15/9-19 BT2 to NOD 15/9-19 B, both 4,250 m; 15/9-19 ST2 to NOD 15/9-19 SR, both 4,641 m; and 15/9-F-11 T2 to NOD 15/9-F-11, 4,562.1 m and 4,562.0 m). The export was retrieved on 24 October 2026. Its rows for the Volve wellbores match an export of 8 September 2026 except for the register's own synchronisation date. All comparisons are against the current register as accessed in October 2026.

In the 15/9-19 series the DDR narratives describe the events behind the designations, and the NOD wellbore histories (NOD 2026b, 2026c, 2026d) are consistent with them. The drillstring in 15/9-19 B became stuck at 3,272 m. A backoff attempt released the string too high, the string was cut at 3,164 m, a sidetrack plug was set between 2,983 and 2,835 m, and new hole was drilled from 2,911 m as 15/9-19 BT2, whose first report reads "REPORTING TRANSFERRED FROM 15/9-19B". Well 15/9-19 S followed a similar sequence. The hole packed off at 2,732 m and attempts to work the pipe free were made at 2,741 m, a cutter was detonated at 2,658 m, the pipe was cut at 2,516 m, three cement plugs were set (the last from 1,543 to 1,430 m), and the narrative uses sidetrack language ("MU BHA FOR SIDE TRACK"). Cement was then drilled out to 1,486 m, and new hole was drilled as 15/9-19 ST2 from a depth between 1,486 m and 1,536 m, the measured depth of the first ST2 report. In both cases NOD records the final total depth against the parent designation.

4 Data Quality

The population of ten WITSML-derived columns was surveyed across all 1,759 DDRs (outputs/stage3_table1_field_population.csv).

The structured kick-off field is populated in only 14.10% of rows (248 of 1,759), so sidetrack kick-off depths had to be recovered from the narratives and from depth sequences (Section 5.7).

A standard null-or-blank check reports 100% population for sum24Hr, but 73 records hold the literal string "None". The string sits in the source XML, in 73 of the 1,759 files, and is a placeholder of the archive. A null test cannot detect it, and the genuine narrative population is 95.85% (1,686 of 1,759). The same placeholder fills 103 forecast24hr values, which leaves a genuine forecast population of 94.14% (1,656 of 1,759). The archive holds 515 placeholder values across all extracted fields, made up of 331 depth sentinels, 7 zero depths, 176 literal "None" strings (73 narratives and 103 forecasts) and one date sentinel.

A character-level survey of sum24Hr found two non-ASCII characters that matter for classification, the one-half sign (½) in 61 DDRs (82 occurrences) and the one-quarter sign (¼) in 10 DDRs (14 occurrences). Both are normalised before pattern matching. The raw files declare the ISO-8859-1 character encoding, and reading them with another encoding turns ½ into Â½ (for example 'due to

double 3-½" DP' for 'due to double 3 ½" DP'). The encoding must therefore be set explicitly. One further anomaly is a single record for 15/9-19 A dated 1 January 1980 (filename 15_9_19_A_1980_01_01.xml, narrative "None", no measured depth), 17.6 years before the first genuine report of the wellbore on 25 July 1997. It is a date sentinel.

5 Results

5.1 Coverage.

The classifier assigns a phase to 1,303 of the 1,759 DDRs, a coverage of 74.1%, and leaves 456 unclassified (25.9%). The largest phases are 12¼ in. and smaller hole (256 reports), Completion (228) and Liner/Casing run (144); all counts are in outputs/table2_phase_distribution.csv. Coverage is the share of reports that match a pattern. It does not measure whether the phase is correct (Section 5.3). The classifier was also checked against an independent earlier extraction of the same archive by the author, with identical results for all 1,759 reports. Fig. 2 shows the classified phases on the depth-versus-time curves of representative wellbores in four panels. The phase colours are single-report labels.

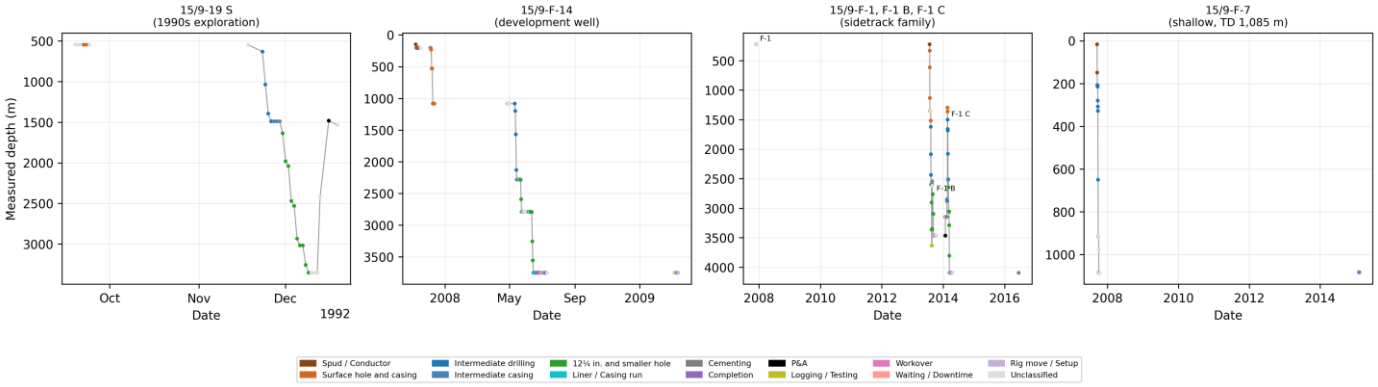


Fig. 2: Depth versus time for six wellbores in four panels (15/9-19 S; 15/9-F-14; 15/9-F-1, F-1 B and F-1 C; 15/9-F-7), coloured by the phase assigned to each report. The phases are single-report labels.

5.2 Stratified Audit and Estimated Recoverable Coverage.

Of the 50 sampled reports, 26 (52%) were labelled construction and 24 (48%) non-construction. The construction group (39 reports) split 23 construction to 16 non-construction, and the production group (11 reports) split 3 to 8. Of the 26 construction reports that the classifier missed, 9 reported genuine drilling without a hole size, 10 described other operations with no corresponding pattern, such as hydraulic fracturing or fishing, and 7 used non-standard terms, such as "CMT" for cement. Weighting the two groups by their share of the 383-report frame gives 51.8% construction (design-based 95% interval 39.1 to 64.5%, normal approximation, with only 11 records in the production group), which corresponds to about 198 reports. The estimated recoverable coverage is therefore 85.3% (95% interval 82.6 to 88.1%): the share of the 1,759 reports that would carry a phase if the patterns also caught the construction content of the unclassified reports. It estimates how much construction content the patterns leave unassigned and does not guarantee that a recovered report would receive the correct phase. The arithmetic, with the Wilson interval for comparison, is in the repository (outputs/audit_arithmetic.md).

The 73 DDRs whose sum24Hr field contains the literal string “None” provide no operational narrative for the classifier used here. They are excluded from the recoverable-coverage estimate. A classifier relying only on each report’s sum24Hr field would therefore leave at least 4.2% of the 1,759 reports unclassified for lack of narrative content. Other information, such as neighbouring reports or additional fields, might support an interpretation, but was not used for this estimate.

5.3 Phase Label Agreement.

The classified phases were checked with a second audit. Four reports were drawn at random from each of the 13 phases, 52 of the 1,303 classified reports, with the seed recorded in the repository. The author, a petroleum engineer, labelled the main operation of each day from the report alone, without seeing the assigned phase, as an engineer would read it. That reading is not the classifier's priority rule. The result is therefore agreement with engineering interpretation and not accuracy against a reference standard.

The frequency-weighted agreement is 61.0%. It weights the agreement rate of each phase (reports agreeing out of four) by the share of the 1,303 classified reports that the phase holds, the sum over the 13 phases of $(N_h / 1{,}303) \times (k_h / 4)$, where N_h is the number of classified reports in phase h. The raw count is 30 of 52 reports (58%). The stratified design-based 95% interval of the weighted agreement is 44.9 to 77.2% (normal approximation, finite-population correction). With four reports per phase, and 4 of the 13 phases at 0 of 4 or 4 of 4, the interval is approximate and fragile. For comparison only, the Wilson interval for 30 of 52, which treats the reports as one simple

random sample, is 44.2 to 70.1% (Wilson 1927). Agreement by phase is in outputs/classified_audit_agreement_summary.json. All four sampled Cementing reports and all four Workover reports disagreed.

The audit identified substantial disagreement between the classifier and the author's single-report labels. Post-hoc inspection suggested incidental keyword matches, differences in phase interpretation and possible dependence on campaign context. The effect of adding neighbouring reports was not tested. The AI tool assigned indicative causes to the 22 disagreements from the narratives: 8 an incidental keyword, 5 campaign context (for example, pulling tubing during an abandonment campaign, or waiting on weather before retrieving a riser), 4 whipstock or window milling that the classifier assigns to Workover and the author labelled Intermediate drilling, 2 casing runs that the size-based phase names place in Intermediate casing and the author labelled Liner/Casing run, 1 a day that mixes waiting and rigging up, and 2 unexplained. The author did not assign the causes. Coverage (Section 5.1) counts the reports that carry a phase, and the agreement indicates how far the assigned phase matches the author's reading of a report on its own. The phase counts and the phase colours in Fig. 2 should be read as single-report labels.

5.4 Timeline Dataset.

The timeline dataset holds 1,421 rows, one for each DDR with a genuine md_m value. Of the 1,759 DDRs, 331 carry the sentinel and 7 the zero value. Each row gives the wellbore identifier, the report date, the days from the wellbore's first report with a recorded depth, the measured depth, the assigned phase and the source filename. Days are counted from the first row with a usable depth in each wellbore.

5.5 Total Depth Validation.

Maximum reconstructed measured depth per wellbore was compared with NOD's total depth for all 26 wellbores. Table 2 lists the special cases and Fig. 3 shows the 24 matched wellbores; all 26 rows are in the repository (outputs/td_validation.csv).

Table 2: Total depth validation, special cases (the full 26-row table is in the repository)

Wellbore (WITSML)	WITSML max MD (m)	NOD TD (m)	Diff. (m)	Outcome	Note
15/9-19 B	3,272.0	n/a	n/a	Unmatched	Parent leg abandoned at stuck pipe; NOD 19 B TD matches 19 BT2
15/9-19 BT2	4,250.0	4,250.0	0.0	Within 0.5 m (by depth)	Technical sidetrack, matched by depth to NOD 15/9-19 B
15/9-19 S	3,353.0	3,580.0	227.0	More than 0.5 m	Archive ends before final drilling
15/9-19 ST2	4,641.0	4,641.0	0.0	Within 0.5 m (by depth)	Technical sidetrack, matched by depth to NOD 15/9-19 SR
15/9-F-11	347.0	n/a	n/a	Unmatched	Parent leg; NOD 15/9-F-11 TD matches F-11 T2
15/9-F-11 T2	4,562.1	4,562.0	0.1	Within 0.5 m (by depth)	Technical sidetrack, matched by depth to NOD 15/9-F-11
15/9-F-15	4,090.0	4,105.0	15.0	More than 0.5 m	TD stated as 4,090 m in the reports; cause of the 15 m difference unresolved

Within 0.5 m: difference of 0.5 m or less. The three technical sidetracks (15/9-19 BT2, 19 ST2 and F-11 T2) have no NOD record of their own and are matched to the NOD record whose total depth agrees. Agreement within 0.5 m is therefore the matching condition for them.

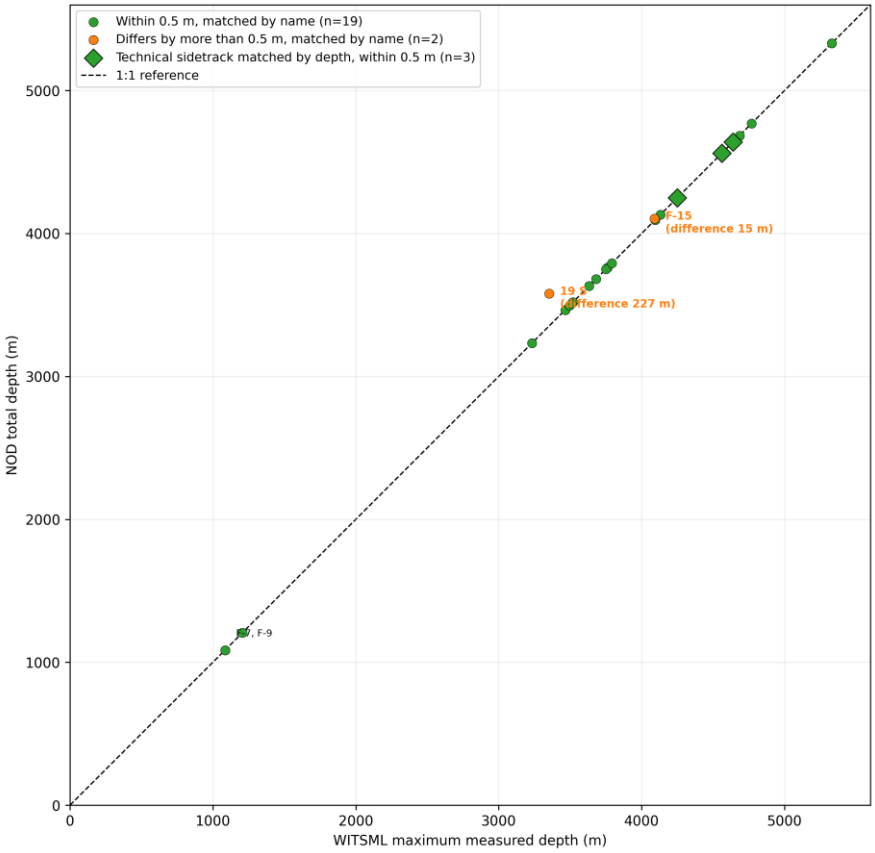


Fig. 3: Reconstructed WITSML maximum MD versus NOD total depth for the 24 matched wellbores, with 1:1 reference line. The two unmatched wellbores (15/9-19 B and 15/9-F-11) are not plotted. Circles mark the 21 wellbores matched by name and diamonds the 3 technical sidetracks matched by depth.

Of the 26 wellbores, 21 are matched to NOD by name, 3 technical sidetracks are matched by depth and 2 parent legs are unmatched. Of the 21 name-matched wellbores, 19 agree on total depth within 0.5 m. 15/9-F-15 differs by 15 m (0.4%) and 15/9-19 S by 227 m. The reports state the total depth of 15/9-F-15 as 4,090 m ("Drilled 8 1/2" hole from 3889m to 4090m (TD)."), NOD records 4,105 m, and the cause of the 15 m difference is unresolved. The 3 depth-matched sidetracks agree by construction of the match.

One explanation covers the two unmatched wellbores and the three depth-based matches. Each WITSML technical sidetrack (19 BT2, 19 ST2, F-11 T2) has no separate NOD entry, and NOD records its final total depth against the parent designation (19 B, 19 SR, F-11). The WITSML parent records that end short of that depth, 15/9-19 B at 3,272 m and 15/9-F-11 at 347 m, are the legs abandoned when the sidetrack was drilled, and they have no NOD counterpart. The two registers differ by construction, and neither is in error.

The WITSML series of 15/9-19 S ends at 3,353 m, 227 m short of NOD's structured total depth (3,580.0 m). NOD's wellbore history records the well as temporarily suspended at 3,569.0 m (NOD 2026d), 216 m deeper than the last WITSML depth. The archive holds no reports for the final drilling and suspension, and the 227 m gap reflects those missing records. The reports it does hold are incomplete throughout, 39 reports over 91 days (Section 5.8), and the records do not show why.

5.6 Spud Date Validation.

Spud date was compared across three sources for every wellbore: the WITSML spud_date field, free-text spud language in the earliest DDRs of the wellbore and the NOD entry date. The offset is the WITSML spud date minus the NOD pre-drill entry date where NOD records one, which marks the start of top-hole drilling, and otherwise the NOD main entry date. Table 3 summarises the outcome and Fig. 4 plots the offset by category. For the four wellbores whose archive contains the pre-drill entry (15/9-F-4, F-5, F-14 and F-15), the first report falls on that date, or one day later for F-4. The categories were first assigned by inspection, and the rules in Table 3 were formalised afterwards to describe those assignments. The final category of 15/9-F-11 is Unresolved: NOD records a pre-drill campaign (6 to 17 April 2012) that is absent from the DDR archive, and the archive's 2013 reports describe a fresh spud from seabed, including the 30-inch conductor. The records do not show how the two campaigns relate. The earlier call of Exact, made by inspection, was not retained.

Table 3: Spud date validation outcomes

Outcome	Wellbores	Rule	Detail
Exact agreement	6	R4: WITSML within 1 day of NOD, and the first report or free-text spud follows by at most 1 day.	15/9-19 A, F-10, F-15 A, F-15 B, F-15 C, F-15 D
Definitional offset	8	R6: within 8 days of NOD, free-text start 2 to 20 days later. R5a: parent leg whose NOD record belongs to its technical sidetrack, with no free-text spud date.	15/9-19 B, F-1 A, F-1 B, F-1 C, F-5, F-9 A, F-11 B, F-14
WITSML data-entry error	4	R2: differs from the first report or free-text spud by more than 30 days, while NOD agrees with them within 3 days.	15/9-F-4, F-7, F-12, F-15
NOD date marks later campaign	1	R3: within 7 days of the first report, more than 1 year from NOD.	15/9-F-1
Inherited parent spud date	2	R1: same spud date as the parent.	15/9-F-11 A, F-11 T2
Unresolved	5	R5b: technical sidetrack whose NOD record is its parent leg's. R7: no other rule fits.	15/9-19 BT2, 19 S, 19 ST2, F-9, F-11 (for F-11 see the text)
Total	26		

Rules are applied in the order R1 (inherited), R5a (parent leg) and R5b (technical sidetrack), R2 (error), R3 (later campaign), R4 (exact), R6 (definitional offset), R7 (unresolved). Unresolved includes 15/9-19 S, F-9 and F-11 (R7).

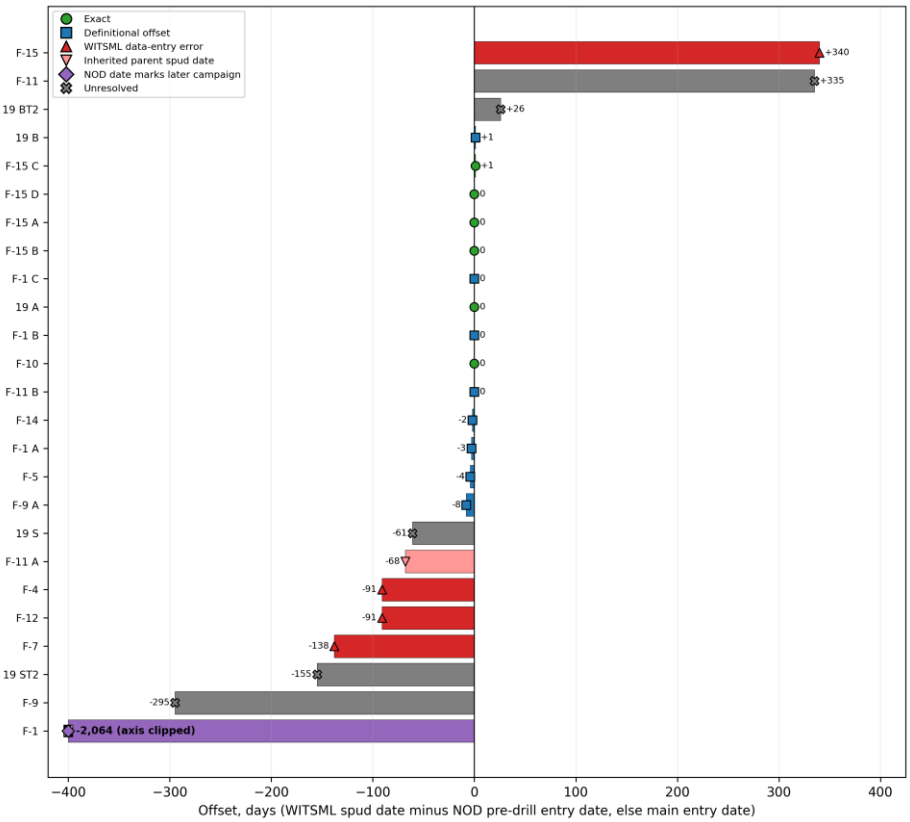


Fig. 4: Spud date offset (WITSML spud date minus the NOD pre-drill entry date where recorded, otherwise the NOD main entry date) for 25 wellbores, coloured by validation category. The bar of 15/9-F-1 (offset -2,064 days) is clipped at -400 days and labelled with its value. 15/9-F-11 T2 is excluded because NOD does not register it separately. A marker at the end of every bar makes zero-day offsets visible, and its shape and colour both identify the category. 15/9-F-11 is shown as Unresolved.

In the four error cases the structured spud_date differs from the first report and its narrative by 91 to 340 days, while the first report and the NOD date agree with each other (for 15/9-F-12, the first report is dated 14 June 2007 and the structured field 15 March 2007). Two sidetracks, 15/9-F-11 A and 15/9-F-11 T2, carry a spud_date identical to that of their parent, which is almost certainly an inherited field value. NOD does not register 15/9-F-11 T2 separately, and Fig. 4 excludes it.

The structured spud_date of 15/9-F-1 (28 November 2007) falls four days before its first DDR (2 December 2007). The 2007 reports cover the top hole only (four reports, maximum depth 226 m), and the well reached its total depth of 3,632 m in the 2013 campaign, whose first report (22 July 2013) is one day before the NOD entry date (23 July 2013). NOD records no pre-drill date for the 2007 work,

unlike F-4, F-5, F-14 and F-15. The NOD date therefore marks the later campaign, and the case is classified as "NOD date marks later campaign". The WITSML value is consistent with the chronology of the wellbore.

5.7 Sidetrack Kick-Off Depth Recovery.

Kick-off depth was recovered from the DDRs because the structured kick-off field is sparse. Kick-off here means the depth at which a sidetrack departs its parent wellbore, not the depth at which an original wellbore begins to build angle; the structured kick-off field of 15/9-19 S (562 m in all its reports) is consistent with the latter and is not used. The depth was read from the last report of the parent and the first report of the sidetrack, in a fixed order of preference: (1) a depth stated for the point where the narrative first reports new formation or departure from the parent hole, or where the kick-off started; (2) for a casing exit by a milled window, the top of the window; (3) where the narrative names the kick-off event but gives no depth, the measured depth of the same report; (4) where the narrative is empty, the measured depth at the parent-to-child transition. The first step with evidence is used. A milled window gives a window interval and no single departure depth. Step 2 therefore applies to it. Confidence follows from the evidence: HIGH where the narrative states the depth together with the departure or the window top, MEDIUM where it states the start of the kick-off or the event without a departure depth, and LOW where the evidence is an interval or the narrative is empty. These definitions describe the grades assigned in Table 4.

Table 4: Sidetrack kick-off depths, special cases (all 15 sidetracks are in outputs/kickoff_table.csv)

Sidetrack	Parent	Kick-off depth (m MD)	Method	NOD kick-off (m)	NOD minus DDR (m)	Confidence
15/9-19 B	15/9-19 A	2,231	depth in text	2,200	-31	HIGH
15/9-19 ST2	15/9-19 S	1,486 to 1,536	depth sequence (narrative empty)			LOW
15/9-F-1 A	15/9-F-1	2,620	event in text, depth from md	2,620	0	MEDIUM
15/9-F-1 B	15/9-F-1 A	2,617	event in text, depth from md	2,604	-13	MEDIUM
15/9-F-1 C	15/9-F-1 B	1,364	depth in text	1,364	0	HIGH
15/9-F-11 T2	15/9-F-11	246	depth in text			MEDIUM
15/9-F-11 B	15/9-F-11 A	2,589	end of stated interval (2,531 to 2,589 m)	2,586	-3	LOW
15/9-F-15 C	15/9-F-15 B	2,566	depth in text	2,575	+9	MEDIUM
15/9-F-15 D	15/9-F-15 C	1,380	event in text, depth from md	1,380	0	MEDIUM

Depths are m MD. NOD minus DDR is NOD's wlbKickOffPoint minus the recovered depth; blank where NOD has no value. For 15/9-19 A, NOD's wellbore history states that the sidetrack was kicked off from 15/9-19 SR at 2,178 m MD. NOD records the 1997 re-entry and plugback preceding that sidetrack as 15/9-19 SR2, a re-entry of 15/9-19 SR. The WITSML archive contains no reports designated 15/9-19 SR2; the kick-off is therefore read from the 15/9-19 A reports and compared with the NOD record. Parent follows the designation chain, each sidetrack being listed under the wellbore designated before it. F-1 B, F-1 C, F-15 C and F-15 D kick off at a depth shallower than the kick-off of that parent and leave the hole shared with earlier wellbores of their family (Section 6.3). For 15/9-19 ST2 the range runs from the depth to which cement was drilled out to the measured depth at the transition, and Table 5 uses 1,536 m. Table 4 lists the sidetracks whose confidence is below HIGH, whose NOD difference exceeds 10 m or whose kick-off is shallower than that of the designated parent; the other six (15/9-19 A, 15/9-19 BT2, 15/9-F-9 A, 15/9-F-11 A, 15/9-F-15 A, 15/9-F-15 B) are in the repository file.

A kick-off depth was recovered for all 15 sidetracks (Table 4 and the repository file). The depth is written in the narrative for 11 of them (step 1 for 9 and step 2 for 2). For 3 (15/9-F-1 A, F-1 B and F-15 D) the narrative names the kick-off event but gives no depth, and the depth is taken from the measured depth of the same report (step 3). For 15/9-19 ST2 both reports at the transition are empty (step 4). The kick-off lies between 1,486 m, the depth to which cement was drilled out in the 15/9-19 S report of 18 December 1992 (the day before the transition), and 1,536 m, the measured depth at the transition. Three cement plugs had been set above it, the last from 1,543 to 1,430 m, and the confidence is LOW. The last report of F-1 B (22 February 2014) reads "Commenced time drilling to1364 [sic] m. 50% formation obtained at 1364 m.", and the first report of F-1 C drills 17½ in. hole from that depth, so the well was sidetracked at 1,364 m. The narratives report 3,059 m on 13 March 2014, but that depth marks the start of the 8½ in. section after casing was run. The sidetrack 15/9-19 A kicked off at the top of a milled window at 2,178 m (window milled from 2,178 to 2,183 m), a depth that NOD also records.

NOD's wlbKickOffPoint is available for 12 of the 15 sidetracks and agrees with the recovered depth to within 10 m for 10, and equals it for 6. The two larger differences are 15/9-19 B (31 m) and 15/9-F-1 B (13 m). The least certain depths are 15/9-19 ST2, which lies between 1,486 and 1,536 m and rests on the measured depth at the transition, and 15/9-F-11 B, where cement and new formation were drilled from 2,531 to 2,589 m and the end of the interval is taken as the departure.

5.8 Duration to Total Depth.

Table 5 gives the duration of the flagged and indicative wellbores and of 15/9-19 ST2; all 26 wellbores, with per-phase report counts, are in outputs/duration_table.csv. Duration runs from the first report (the 1980 date sentinel excluded) to the date total depth was first reached, which is the maximum valid depth. The table lists the elapsed calendar days, the number of reports in that span (inclusive), the drilled length (total depth minus the recovered kick-off depth for the 15 sidetracks, Section 5.7, and total depth for the other 11) and

metres per report, which is the drilled length divided by the number of reports. With inclusive counting, the number of reports equals the elapsed days plus 1 for a wellbore reported daily, and falls short of that where the archive has gaps. Seven of the 26 wellbores, whose elapsed days exceed twice their reports, are flagged (15/9-19 S, F-1, F-4, F-5, F-9, F-14 and F-15). F-1 reaches its total depth of 3,632 m in 31 reports spread over 2,085 days, four of them from 2007. The totals of 15/9-19 B (3,272 m) and 15/9-F-11 (347 m) are the depths of an abandoned parent leg or stub (Section 5.5) and make their durations indicative only. Metres per report is not a rate of penetration. It includes non-drilling days, gaps between reports and, for F-1, two campaigns.

Table 5: Duration to total depth (TD) and drilled length, flagged and indicative wellbores and 15/9-19 ST2 (all 26 wellbores are in outputs/duration_table.csv)

Wellbore	TD (m)	Elapsed days to TD	Reports to TD	Gap flag	Drilled length (m)	m per report
15/9-19 B (indicative)	3,272	18	19		1,041	54.8
15/9-19 S	3,353	91	39	yes	3,353	86.0
15/9-19 ST2	4,641	83	68		3,105	45.7
15/9-F-1	3,632	2,085	31	yes	3,632	117.2
15/9-F-11 (indicative)	347	12	13		347	26.7
15/9-F-14	3,750	222	61	yes	3,750	61.5
15/9-F-15	4,090	376	55	yes	4,090	74.4
15/9-F-4	3,510	251	40	yes	3,510	87.8
15/9-F-5	3,792	258	31	yes	3,792	122.3
15/9-F-9	1,085	275	14	yes	1,085	77.5

Gap flag: elapsed days to TD exceed twice the reports to TD. Drilled length is total depth minus the recovered kick-off depth for sidetracks and total depth for original wellbores; for the original wellbores this is measured depth from the rotary kelly bushing (RKB) and includes the water depth. (indicative): TD is the depth of an abandoned parent leg or stub. For 15/9-19 ST2 the kick-off used is 1,536 m, the deeper end of the range of 1,486 to 1,536 m (Section 5.7), so its drilled length is a minimum (3,105 m, or 3,155 m at 1,486 m, which gives 45.7 to 46.4 m per report).

6 Discussion

6.1 What the Dataset Supports.

The 1,421-row depth-time-phase dataset can be queried directly. The classified phases locate casing, cementing and P&A operations in time and depth for each wellbore as single-report labels. Phase counts and durations should not be used directly for offset-well planning, because the phases are single-report labels that agree with the author's labels for 61.0% of reports weighted by phase size (Section 5.3) and the archive has gaps. The residual audit estimates the aggregate amount of construction content left unassigned. It does not establish phase-specific completeness or bound errors in phase counts and durations. The estimate is roughly 198 reports, mostly drilling progress reported without a hole size. Table 5 gives elapsed days, reports and metres per report to total depth for the flagged and indicative wellbores. Metres per report is not a rate of penetration. It includes non-drilling days, gaps between reports and, for 15/9-F-1, two campaigns. The gap flag marks wellbores whose elapsed time cannot be read as drilling time.

6.2 Data-Quality Lessons for WITSML Archive Users.

The characteristics of the archive that would mislead an analysis of the structured fields alone (placeholder values that pass null tests, a date sentinel, structured spud dates that contradict the first report, a sparsely populated kick-off field, and technical-sidetrack designations without a register counterpart) are described in Section 4 and in Sections 5.5 to 5.7. Each needs an explicit rule, and the rules used here are published with the code.

6.3 Relevance to Records-Based Screening of Legacy Wellbores.

Assessments of depleted fields for CO2 storage, and decommissioning reviews generally, rely on an inventory of existing wellbores and on evidence of how each was constructed and abandoned. The reconstruction shows that a national register alone gives an incomplete inventory in three ways. Abandoned parent legs below technical sidetracks (15/9-19 B to 3,272 m and 15/9-F-11 to 347 m) exist physically but have no register entry of their own. A sidetrack shares the hole of its parent above its kick-off depth. The wellbores of one family are therefore not independent penetrations above that depth. The 15/9-F-1 family shares the hole above 1,364 m MD, the five wellbores of the 15/9-F-15 family share it above about 1,380 m MD, and 15/9-19 S, ST2 and A share it above the kick-off of each sidetrack (In the 15/9-19 family, the interpreted 15/9-19 ST2 departure from 15/9-19 S lies between 1,486 and 1,536 m MD. NOD describes 15/9-19 A as kicked off from 15/9-19 SR at 2,178 m MD, following the 1997 re-entry and plugback designated 15/9-19 SR2). The archive can also stop, as it does for 15/9-19 S, which has no reports covering its final 227 m. A screening assessment should record such a gap as an evidence gap without inferring the missing history. The phase-classified timeline is a starting point for locating the casing, cementing and plugging records that such an assessment needs.

6.4 Transferability.

The audit protocol, the placeholder checks and the wellbore-matching logic can be applied to WITSML 1.4-series drillReport archives after adaptation and validation (tested here on 1.4.0.0). The pattern set is tuned to the English-language narratives and abbreviations of one operator over the life of one field. Applying it elsewhere requires a pattern review. The audit should be repeated to measure the new residual, since the Volve figure cannot be assumed to carry over.

7 Limitations

The classifier assigns one primary phase to each DDR, using that report alone. A DDR that narrates both drilling progress and cementing takes the higher-priority drilling phase, and the Cementing count therefore underrepresents the days on which cementing occurred alongside another operation. The agreement audit (Section 5.3) compares the classifier with one person's reading of single reports, did not test neighbouring reports, and relies on causes of disagreement that the AI tool assigned from the narratives.

The residual audit labelled construction content only, which makes the recoverable coverage conservative for non-construction phases such as waiting or rig moves, and it does not guarantee that a recovered report would receive the correct phase. One reviewer, the author, labelled the 50 records. The design-based interval of 82.6 to 88.1% rests on a normal approximation, which is optimistic with 11 records in one group. Two sampled records sit in a group that does not match their context: the 1980 date sentinel places the 1997 drilling of 15/9-19 A in the production group, and the 2013 campaign of 15/9-F-1 also falls there.

The timeline holds only DDRs with a genuine measured depth and applies no interpolation. A discontinuity marks missing data and supports no conclusion about drilling activity in that period.

Two kick-off depths rest on weaker evidence: 15/9-19 ST2 on the measured depth at the transition (between 1,486 and 1,536 m) and 15/9-F-11 B on an interval of cement and new formation (2,531 to 2,589 m). NOD gives no kick-off point for three sidetracks. Wellbores of one family share the hole above their kick-off depths (Section 6.3), and any formation-level analysis must account for this. Metres per report are not rates of penetration, and gaps between reports affect the elapsed time to total depth (Table 5).

No isolated ablation was run for Unicode normalisation (the one-half sign in 61 DDRs and the one-quarter sign in 10). An extraction error would propagate to every result, and the agreement with NOD on total depth (19 of 21 name-matched wellbores within 0.5 m) indicates that the depth fields were extracted faithfully.

8 Conclusions

The Volve case shows that reconstructing a well’s construction history is a records-reconciliation problem as well as a classification problem. Structured fields provide a useful starting point, but placeholder values, inherited dates, incomplete reporting and differences in wellbore designation require examination of the underlying narratives. Comparison with NOD records supports the reconstructed maximum depths for most name-matched wellbores, while the remaining discrepancies are retained explicitly rather than forced into agreement.

The phase classifier provides a transparent and reproducible way to organise the reports, but its labels should not be treated as verified operational phases. The agreement audit demonstrates substantial differences between the classifier’s priority rules and the author’s interpretation of the main operation. The principal contribution is therefore an auditable reconstruction of the available archive, with documented evidence gaps and interpretation limits, not a complete operational history or an automated well-integrity assessment. For legacy-well screening, the practical value is a traceable route from a wellbore inventory to the reports needed for further engineering assessment.

- Reproducible reconstruction: the archive, wellbore population and 1,421-record output.
- Classification performance and limits: 74.1% coverage, 61.0% weighted agreement, and the distinction between the two.
- External comparison: 19/21 name-matched depth agreements, with depth-assisted matches and unresolved discrepancies reported separately.
- Records reconciliation: placeholders, spud-date problems and technical-sidetrack representation.
- Engineering application: a starting point for locating and assessing construction evidence, not proof of well integrity.

Further work should establish a common phase-labelling protocol with independent reviewers, evaluate classification using neighbouring reports and campaign context, and compare the reconstruction with Final Well Reports and End of Well Reports. These steps would test whether the dataset can support more detailed operational analysis beyond the records-based screening demonstrated here.

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Generative AI (a commercially available generative AI assistant based on a large language model) built and ran the reproducible analysis pipeline from the raw data and was used to assist with debugging, checking arithmetic and editing the manuscript text. The sidetrack kick-off depths were proposed by the AI tool from the narratives under a fixed order of preference. The author rejected the tool's first proposals for three sidetracks (15/9-19 A, 15/9-F-9 A and 15/9-F-11 B) and kept the earlier values, then reviewed all 15 depths and approved every one without change. The author made all audit labels (50 in the audit of unclassified reports and 52 phase labels in the agreement check), reviewed all AI-assisted content, and takes full responsibility for the paper. The causes of the 22 disagreements in the agreement check were assigned by the AI tool from the narratives and are indicative.

Code and Data Availability

The classifier, the extraction and analysis code, the 1,421-row timeline dataset, the audit labels (file name, wellbore, date, stratum, call and miss reason), the phase-agreement labels (record id, wellbore, date, assigned phase and the author's phase, with the category and proposed cause of each disagreement; no narrative text), the NOD comparison tables, the kick-off and duration tables, and a numbers registry tracing every reported figure to the script that produced it are available at https://github.com/985185/volve-well-construction-timeline. DDR narrative text is not redistributed: the Volve data are available from Equinor under the Equinor Open Data Licence (https://www.equinor.com/energy/volve-data-sharing), and the data files of the repository hold only file names, dates, depths, categories and quotes of 15 words or fewer. The check against the author's earlier extraction (Section 5.1) is reported but is not needed to reproduce any figure. Agreement between the classifier and the author's labels can be recomputed from the published paired labels, but checking the labels themselves requires the source narratives from the Equinor dataset. Every quantitative figure in this paper traces to an entry in the numbers registry published in the repository (307 entries, each with the producing script and function, the SHA-256 of its input files and an execution timestamp). The audit estimates in Section 5.2 are derived arithmetically from registry entries, and the arithmetic is shown in outputs/audit_arithmetic.md.

Nomenclature

BHA = bottom-hole assembly
BOP = blowout preventer
DDR = Daily Drilling Report
DP = drill pipe
LWD = logging while drilling
MD = measured depth
MWD = measurement while drilling
NOD = Norwegian Offshore Directorate (formerly Norwegian Petroleum Directorate)
P&A = plug and abandonment
RKB = rotary kelly bushing
TD = total depth
VSP = vertical seismic profile
WITSML = Wellsite Information Transfer Standard Markup Language
WOW = waiting on weather

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SI Metric Conversion Factors

To convert	Multiply by	To obtain
in.	2.54 (exact)	cm
sg (relative density, water = 1)	1,000 (exact)	kg/m3

Hole and casing sizes in this paper are nominal inch designations as written in the reports (for example 12¼ in. hole and 9⅝ in. casing), and depths are in metres (m).